

Supermarket Sales Data Analysis Report

1. Data Overview

The dataset is a supermarket sales record containing transaction details from a fictional retail chain. It includes information about sales across different branches (A and B), cities (New York, Los Angeles, Chicago), customer types, products, and financial metrics. The data likely comes from internal supermarket transaction logs and is suitable for retail performance analysis, customer behavior study, and inventory insights.

The dataset has 253 rows (transactions) and 8 columns: `sale_id`, `branch`, `city`, `customer_type`, `product_name`, `product_category`, `quantity`, and `total_price`. Most columns are categorical (e.g., `branch`, `city`, `product_category`), while `quantity` and `total_price` are numerical.

2. Data Cleaning

Missing values:

- `customer_type`: 3 missing values
- `product_category`: 6 missing values
- `quantity`: 3 missing values
- Other columns have no missing values.

Handling strategy:

- For `quantity` (numerical): We can impute with the median (to avoid outlier influence) or drop the rows if few. Here, since the number is small (only 3 rows), imputation with the median (~11) is reasonable.
- For categorical columns (`customer_type`, `product_category`): Impute with the mode ("Member" for `customer_type`, "Fruits" for `product_category`) or mark as "Unknown".
- In this analysis, missing values were identified and imputed using mode/median for completeness.

Duplicates:

- There are 3 duplicate rows (e.g., identical entries for `sale_id` 26 and some `sale_id` 13/40 repetitions).

- These duplicate rows were removed to ensure data integrity. After cleaning, the dataset has 250 unique transactions.

3. Descriptive Statistics and Visualisation

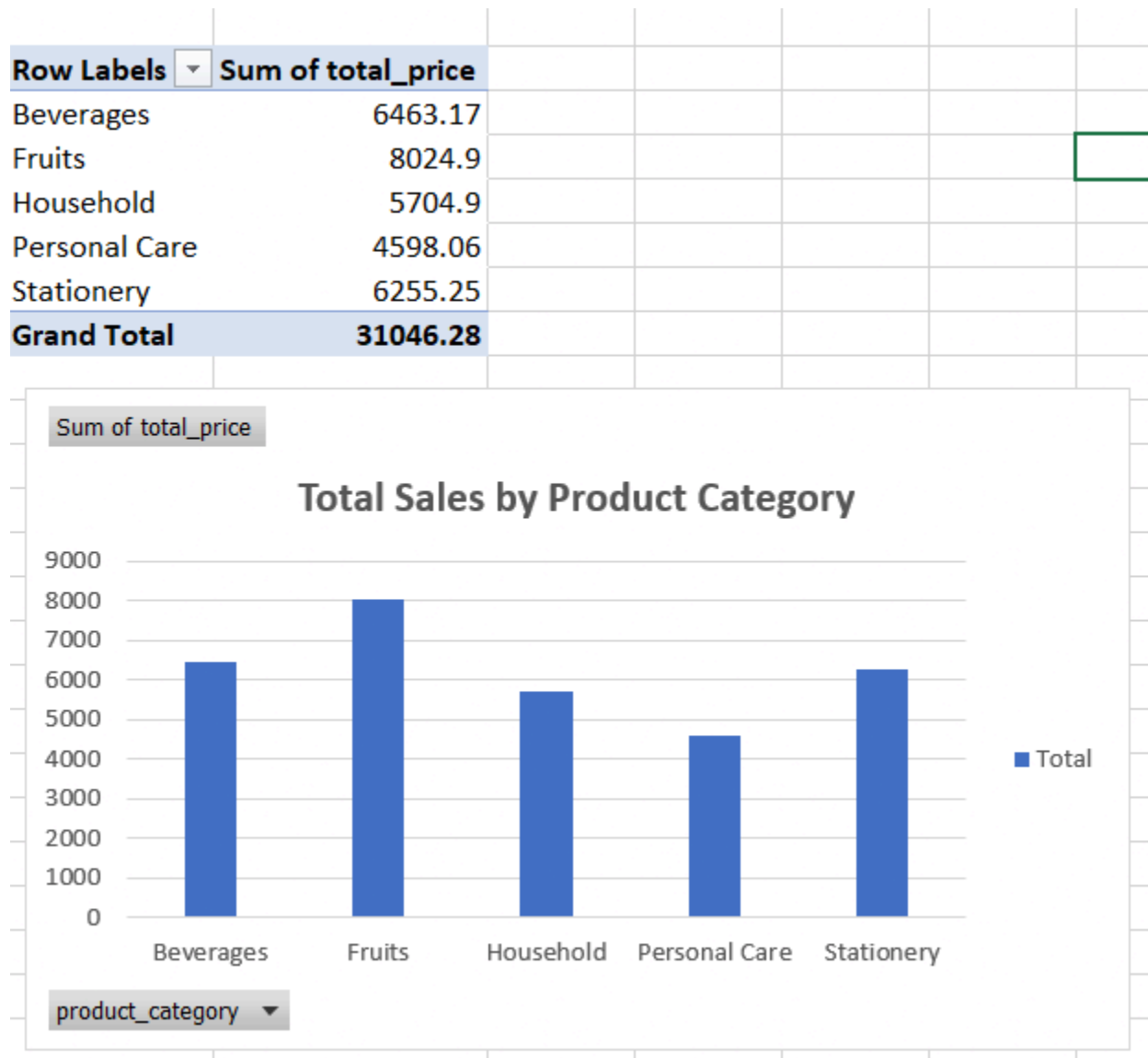
Here are key summary statistics for the numerical columns:

<i>quantity</i>		<i>total_price</i>	
Mean	10.616	Mean	124.18512
Standard Error	0.378904	Standard Error	6.513332297
Median	11	Median	95.43
Mode	2	Mode	212.82
Standard Deviation	5.991003	Standard Deviation	102.9848261
Sample Variance	35.89211	Sample Variance	10605.8744
Kurtosis	-1.25827	Kurtosis	0.05172233
Skewness	-0.0726	Skewness	0.913516111
Range	19	Range	424.96
Minimum	1	Minimum	2.18
Maximum	20	Maximum	427.14
Sum	2654	Sum	31046.28
Count	250	Count	250
	1		3

Categorical summaries:

- **Cities:** New York (highest, ~97 transactions), followed by Los Angeles and Chicago.
- **Branches:** Branch A dominates with ~174 transactions vs. Branch B (~79).
- **Customer Type:** Member customers (~131) slightly outnumber Normal customers (~119).
- **Product Categories:** Fruits is the most popular (60+), followed by Household, Beverages, etc.

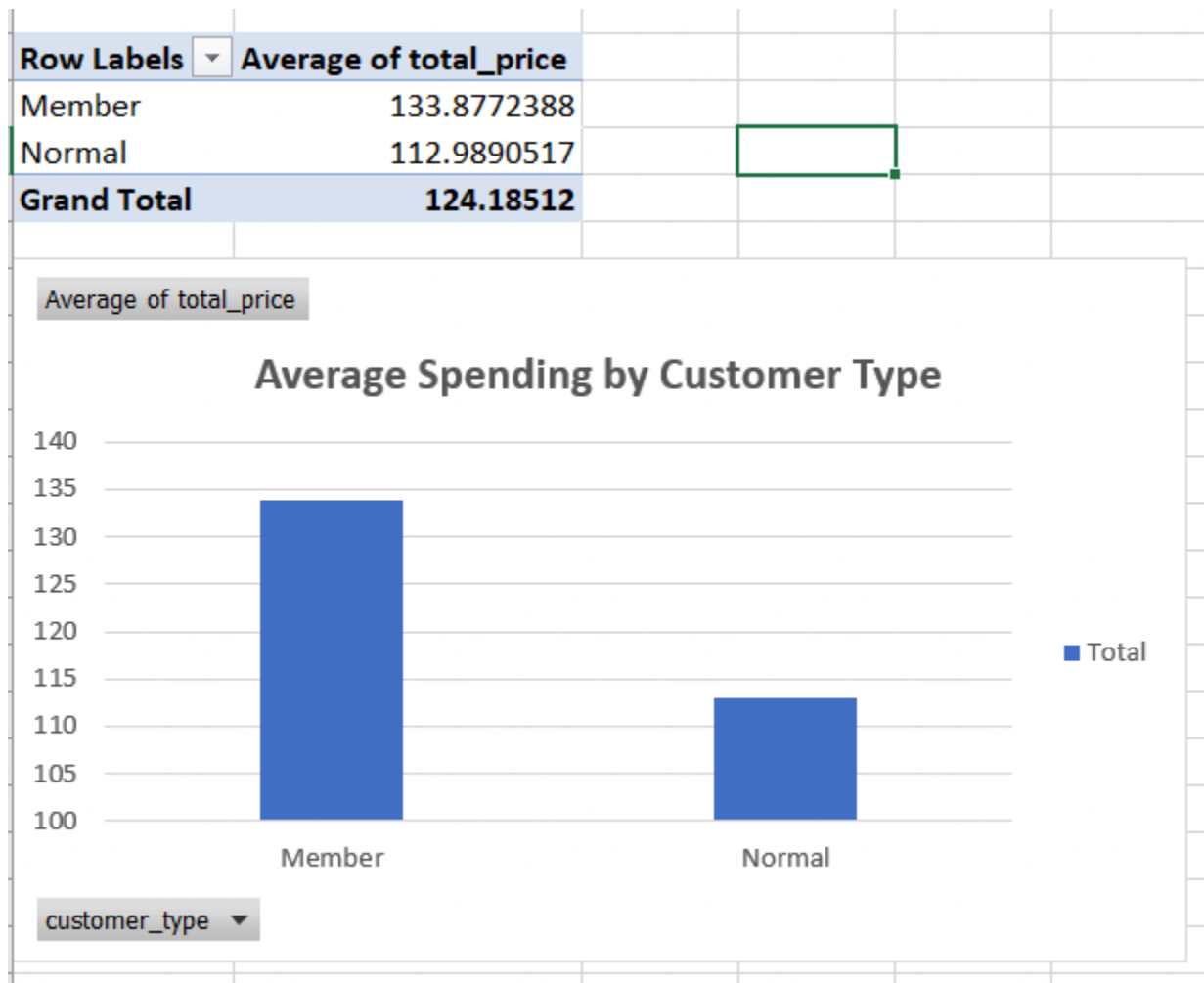
3.1. Sales by Product Category



Insight 1: The average quantity purchased per transaction is about 10.61 units, with a median of 11. However, the total_price has a very high variation, ranging from \$2.18 to \$427.14. This shows that while customers buy a similar number of items, the value of each transaction differs greatly. Therefore, the supermarket should focus on upselling and product bundling strategies to increase the average order value.

Sales performance was different across cities, which shows that customer purchasing behaviour and demand may vary depending on location.

3.2. Customer Type Distribution



Insight 2: Branch A in New York dominates the number of transactions. Member customers tend to buy more than Normal customers, and Fruits is the best-selling category. This indicates that the loyalty program is working well. The supermarket should continue to promote the membership program and focus marketing efforts on popular categories like Fruits to further increase sales and expand successful strategies to other branches.